

Convex

Unique

Unique



**“recovers” pointwise
geometry**

Properties

$$\nu^\star = \arg \min_{\nu} \sum_{i=1}^K \lambda_i W_p^p(\mu_i, \nu)$$

- **Convex** in weights $\{\lambda_i\}$.
- **Unique** under mild conditions (especially with W_2)
- Reduces to:
 - Euclidean mean if: $p = 2, \mu_i = \delta_{x_i}, \lambda_i = \frac{1}{K}$: $\nu^\star = \delta_{\text{mean}(x_i)}$
 - Median if $p = 1, \mu_i = \delta_{x_i}, \lambda_i = \frac{1}{K}$: $\nu^\star = \delta_{\text{median}(x_i)}$
- For $p=2$, mean of barycenter = barycenter of mean
$$\int_X x \, d\nu^\star(x) = \sum_s \lambda_s \int_X x \, d\mu_s(x)$$
- For Gaussians (with W_2): barycenter is also Gaussian

OT geometry again
“**recovers**” **pointwise geometry** in corner cases!

Example: Gaussians